

RoR Research Log

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Chronological record of substantive decisions, rationale, and results for the “Influence of Reviewer Expertise and Engagement on Peer Review of Grants” project. This is the actual decision log for the project – edited by Sam, and by Claude when asked. For a lightweight session-continuity summary (where things stand, what to pick up next), see `PROJECT.md` at the repo root instead; that file is not a substitute for this one.

2026-08-03

Proposal docx converted to Quarto

Converted `writing/cihr-ror-proposal.docx` (a downloaded Google Doc believed to be the source of the funded proposal) into `writing/cihr-ror-proposal.qmd`, matching the repo's existing typst/Quarto template. All 38 numbered in-text citations were mapped to existing `@keys` in `writing/ror.bib` – no new bibliography entries were needed, every key already existed. The Gantt-style timeline figure was extracted from the docx and saved to `writing/media/timeline.png`. Rendered cleanly via `quarto render --to typst` with zero citation warnings.

Typst rendering polish

Two aesthetic fixes to `writing/cihr-ror-proposal.qmd`:

- **Title font mismatch.** The document title was rendering in Quarto's default typst template font (“libertinus serif”) instead of the body font (C059), because the title block in `typst-template.typ` is styled separately from ordinary headings via a `heading-family` parameter that only gets set through the `brand.typography.headings.family` metadata path – not through `mainfont`. Fixed by adding:

```
brand:
  typography:
```

```
headings:
  family: C059
```

- **Section headings too large/bold for this font.** Added an include-in-header block with explicit `show_heading.where(level: N):` rules to size down H1/H2 slightly (1.25em / 1.1em instead of Typst's default ~1.4em / 1.2em).

Fixed brms family-name bug: `student_t()` -> `student()`

User hit Error in `student_t()` : could not find function "student_t" when sourcing `code/ror-analysis-score-models.R`. `student_t()` is a Stan distribution function name, not an R function brms exports – the correct brms family constructor is `student()`. Fixed the `family` = argument and the two comments referencing it.

Outcome choice: model d_{ijk} , not final score conditional on consensus

Question raised: should the primary outcome be the consensus-to-final difference score (d_{ijk} , currently PROJECT.md's stated choice) or the final score modeled directly with consensus score as a covariate? Decided to keep d_{ijk} , for three reasons:

1. **The point mass doesn't disappear if you model final score instead – it just stops being a fixed reference.** A non-deviating member's final score equals consensus exactly. Modeling `final_score ~ consensus_score`
 - ... still has that spike, just located at a value that moves application-by-application (wherever that application's consensus happened to land), instead of a single common value (0). Differencing first standardizes the spike location across every application, which is what makes a shared bernoulli-hurdle-style family tractable.
2. **Avoids reintroducing a confound.** A freely-estimated slope on consensus in a `final_score ~ consensus + ...` model has to recover something close to 1 from the data; if reviewer composition happens to correlate with consensus level across committees, that estimation can leak into the coefficients of actual interest. Differencing with an implicit slope of exactly 1 (which is the true administrative rule, not an assumption) removes that channel by construction.
3. **Matches the actual research question.** Aim 1 is about the *change* induced by discussion, not the *level* of the final score (which is mostly driven by application quality, not the process under study). Nothing is lost: `final_score = consensus + d_ijk` is a trivial identity, so anything Aim 3 needs at the score-level for funding-decision simulations can be reconstructed downstream.

A naive linear model on d_{ijk} has ~zero power to detect known effects

Empirical check on `data/sim-deviation-data.csv` (which has known ground truth – `job/exp` are coded to affect $P(\text{deviate})$ but not the *mean* deviation given deviation, since `dev_bias = 0` in `ror-sim-deviation.R`):

```
> m <- lm(deviation ~ job + exp, data = d)
      Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.0016      0.0016 -0.9956  0.3195
```

```

jobreviewer -0.0014 0.0024 -0.5833 0.5597
explov      0.0040 0.0025 1.5784 0.1145
expmed      0.0017 0.0020 0.8430 0.3992
expnone     0.0052 0.0025 2.0649 0.0389

> m2 <- glm(deviated ~ job + exp, data = d, family = binomial())
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.9202      0.0337 -27.2977    0
jobreviewer -0.2601      0.0484  -5.3764    0
explov       0.6631      0.0497  13.3402    0
expmed       0.4244      0.0410  10.3555    0
expnone      0.8639      0.0495  17.4691    0

```

The linear model's coefficients are tiny and mostly non-significant (largest effect size 0.005 points on a variable bounded at +/-0.5). The logistic model on the same covariates recovers the true, large, highly significant effect pattern baked into the simulator. This isn't a subtle power difference – because deviation magnitude is symmetric around zero given deviation, essentially all covariate signal lives in *whether* someone deviates, and a model that only looks at the conditional mean of d_{ijk} structurally cannot see it. Confirms the two-part (bernoulli x magnitude) model structure is necessary here, not over-engineering.

Simulator bug: scores weren't rounded to the CIHR-mandated tenth

Sam flagged: CIHR reviewer scores (consensus and individual final scores) can only be entered to one decimal place. `ror-sim-deviation.R` was not enforcing this – consensus/score were raw floating-point draws (e.g. 3.808046), not multiples of 0.1.

Fixed by adding a `round_tenth()` helper and rounding consensus, deviation, and score to the nearest tenth. This introduced a correctness wrinkle: rounding a continuous deviation draw can collapse it to exactly 0, which is a contradiction for a row where `deviated == 1` (that member is defined to have deviated). Fixed with rejection sampling – redraw from `rtruncnorm()` and re-round, looping until no `deviated == 1` row has a zero deviation. With `dev_sd = 0.15`, about 26% of raw draws land in the collapse-to-zero band, so this loop does real work (average ~1.35 draws/row), not just handling a rare edge case:

```

> ptruncnorm(0.05, a=-0.5, b=0.5, mean=0, sd=0.15) - ptruncnorm(-0.05, a=-0.5,
b=0.5, mean=0, sd=0.15)
[1] 0.261

```

Added `stopifnot()` checks confirming consensus/deviation/score are all exact tenths and that no `deviated == 1` row has zero deviation. Current distribution of deviation magnitude among deviators (n = 6756 deviators out of 18000 rows):

```

-0.5 -0.4 -0.3 -0.2 -0.1  0.1  0.2  0.3  0.4  0.5
   8   69  333 1026 1915 1952 1033  340   75    5

```

Magnitude submodel switched from `student()` to `ordinal cumulative()`

Direct consequence of the rounding fix above: deviation magnitude among deviators is not continuous – it's one of 10 discrete, unevenly-spaced values. Discussed the trade-off explicitly before changing anything:

- A continuous family (`student()/gaussian()`) assigns probability to impossible values, can't represent the actual (very uneven) spacing between categories – sharply peaked near ± 0.1 , sparse at ± 0.5 – and has no natural bound at ± 0.5 without truncation hacks. The sparse tails are exactly the large swings most relevant to Aim 3's funding-decision question, so a poorly-fit tail is a real cost, not a rounding curiosity.
- An ordinal `cumulative()` model matches the discretization exactly, gets the ± 0.5 bound for free, and (with `threshold = "flexible"`, the default) doesn't assume the 10 steps are equally likely. Cost: contrasts in score-point units require an explicit probability-weighted sum over category values rather than falling out of a default linear prediction; 9 threshold parameters instead of 1 scale parameter (irrelevant at $n \sim 6700$).

Decided to implement it. Changes to `code/ror-analysis-score-models.R`:

- `deviation` recoded to an ordered factor over the 10 signed tenths (matched via `sprintf("%.1f", ...)` string comparison to avoid floating-point equality issues between the simulator's rounding and a plain numeric `factor()` call).
- `m1_magnitude` now `cumulative(link = "logit", threshold = "flexible")` instead of `student()`.
- Priors switched to match the logit scale (`normal(0, 1.5)` on thresholds, `normal(0, 0.5)` on betas, matching `m1_deviate`'s own priors since both models are now logit-linked); dropped the sigma prior (no residual-SD parameter in an ordinal model).

Two follow-ups flagged in the script's TODO block rather than resolved now: (1) the $E[\text{deviation}] = P(\text{deviate}) * E[\text{deviation} | \text{deviate}]$ combination step needs $E[\text{deviation} | \text{deviate}]$ computed as a probability-weighted sum over the 10 category values, not a linear prediction – needs an explicit `marginaleffects/tidybayes` contrast; (2) the 10-category $\pm 0.1 \dots \pm 0.5$ grid is still an assumption carried over from the simulator's parameters, not yet confirmed with CIHR's Funding Analytics Team.

Confirmed: reviewers can deviate from their own consensus score

Sam clarified the DGP: the 3 assigned reviewers must *agree* on a consensus score, but are not obligated to *use* it as their own final score – they can deviate from it just like non-reviewing panelists. Checked `ror-sim-deviation.R`: this was already correctly implemented. `p_dev = plogis(a0 + a1*panelist + ...)` only adds the `a1` term for panelists; reviewers still get a nonzero baseline `p_dev`. Confirmed empirically – reviewers deviate 32.7% of the time in the simulated data, not 0%. Consistent with the early proposal draft's own language (`writing/proposal.qmd`: “all committee members (including each of the reviewers) may assign their final score within a range of ± 0.5 from the consensus score”). No code change needed.

This log created

Split the project's actual decision/rationale record out from `PROJECT.md`. `PROJECT.md` will stay as a lightweight, Claude-oriented continuity note (where things stand, what to pick up next session) – it is not a substitute for a real research log and was never intended as one. This file is that log: chronological, decisions + rationale + evidence, editable by Sam directly and by Claude on request. Mirrors the format used in the `u2-sibs` project's `code/u2s-research-log.Rmd`, adapted to Quarto.